User Guide

Ordinal-Probit Latent-Class Trajectory Model

File: `ordinal_probit.sas` Traj2 (released core) Authors: Haiqun Lin, Anum Zafar

What it does

This file fits a class-specific cumulative-probit trajectory model for a bounded ordinal outcome, such as an ADL score, a frailty grade, or a cognitive screening category, by maximum likelihood in PROC NLMIXED. It sorts subjects into a few latent classes, each with its own trend over time, and each class gets its own ordered thresholds. The thresholds are built from exponential increments, which keeps them ordered without any constrained optimization. Everything is macro-only with no compiled code, so it runs as-is in the CMS VRDC and any SAS 9.4 or newer session.

What you need

- SAS 9.4 or newer with SAS/STAT (PROC NLMIXED).
- ODS Graphics for the plots (`%ordprob_mix_plot_one`). The fit itself runs without them.
- A wide, one-row-per-subject dataset, and write access to WORK.

The model in brief

For an outcome with m ordered categories, each class c has a latent index that is a polynomial in time, and its own thresholds. The probability of each category comes from differences of the normal CDF:

```
eta_c(t) = beta_c0 + beta_c1*t + ... + beta_cd*t^d

thresholds (exp increments keep them ordered):
  th1 = i_c1
  thj = th(j-1) + exp(i_cj),   j = 2 .. m-1

P(Y=1 | c) = probnorm(th1 - eta)
P(Y=j | c) = probnorm(thj - eta) - probnorm(th(j-1) - eta)
P(Y=m | c) = 1 - probnorm(th(m-1) - eta)

class share:  pie_c = exp(alpha0_c) / sum_k exp(alpha0_k)
```

For identification the class A mixing logit is fixed to 0, and the first threshold anchors the location of each class index while its intercept beta_c0 is free. Exp increments guarantee ordered categories with no bounds needed. Missing time points are allowed and skipped under missing-at-random.

Quick start (simulated data)

The simulator, the base build, the fit, and the plot run as four calls. The simulator's cap sets the highest category, so use cap = m - 1 to match your category count.

```
/* 1. Simulate a 4-category outcome (cap=3 gives values 0..3) */
%sim_data(dist=BIN4, class=3, n=2000, T=12, seed=1, cap=3);

/* 2. Build BASE_FILE_SRS (adds quar1..quar12 = 0..11, id -> BENE_ID) */
%build_base_from_simwide(T=12);

/* 3. Fit one K=3, quadratic model */
```

```
%ordprob_mix_fit_one(
  data=BASE_FILE_SRS, id=BENE_ID,
  yvars=Y1_1-Y1_12, tvars=quar1-quar12, ttotal=12,
  m=4, ycodes=0 1 2 3, deg=2, k=3, prefix=ordprob_T1C3);

/* 4. Plot class proportions + predicted trajectories */
%ordprob_mix_plot_one(outestlib=work, prefix=ordprob_T1C3,
  k=3, ttotal=12, m=4, ycodes=0 1 2 3);
```

The settings you edit

Argument	Meaning
`data` / `id`	Input wide table and subject id (defaults `BASE_FILE_SRS`, `BENE_ID`).
`yvars`	Ordinal outcome columns, one per time point (e.g. `Y1_1-Y1_12`).
`tvars`	Time-index columns holding values 0..T-1 (e.g. `quar1-quar12`).
`ttotal`	Number of time points.
`m` / `ycodes`	Number of categories and their codes (e.g. m=4, `0 1 2 3`).
`deg`	Trend shape: 1 linear, 2 quadratic, 3 cubic.
`k`	Number of latent classes.
`start_values`	Optional starting values (see below). Blank starts everything at 0.
`bounds`	Optional parameter bounds for stability.
`prefix`	Prefix for the output dataset names.

Using your own data

Point data= at a wide table with one row per subject, or build BASE_FILE_SRS yourself. It must contain the columns named in your arguments.

Column	What to supply
Subject id	Named by `id`, unique per subject.
Outcome	Named by `yvars`, one integer-coded column per time point.
Time index	`tvars` (quar1-quarT) holding the values 0..T-1. Create these yourself.
Categories	Set `m` and `ycodes` to match your outcome's codes exactly.

Two things to get right:

- Codes must match. A value that is not in `ycodes` is not counted at that time point, so align `ycodes` and `m` with your data.
- Time is 0-based. The build and the plot use t = 0..T-1, so fill `quar1-quarT` with 0..T-1, not 1..T.

Running and choosing the number of classes

Each call to `%ordprob_mix_fit_one` fits one model for a fixed k. To choose the number of classes, refit with different k values and compare BIC in the fit-statistics dataset, preferring the lowest BIC with sensible class sizes and clear separation.

Starting values

Leave start_values blank to start every free parameter at 0. To set your own, pass name=value pairs; the macro merges them with the defaults into one PARMS statement with no duplicates. Do not include the class

A mixing logit (alpha0_A is fixed to 0). Following the data dictionary, supply the threshold increments rather than the first threshold.

```
%let STARTS = %str(alpha0_B=-0.5 alpha0_C=-0.2
                   beta_a0=0 beta_a1=0 beta_a2=0);
%ordprob_mix_fit_one(... start_values=&STARTS ...);
```

What you get back

- `{prefix}_EST_K{k}`: parameter estimates and standard errors (alpha, beta, threshold increments).
- `{prefix}_FIT_K{k}`: fit statistics (AIC, BIC, -2 log L) for comparing candidate k.
- `{prefix}_ESTS_K{k}`: thresholds on the natural scale.
- Plots from `%ordprob_mix_plot_one`: a class-proportion barplot and predicted E[Y_t | class] trajectory curves.

Troubleshooting

Symptom	Fix
Values seem ignored / odd likelihood	Outcome codes do not match ycodes. Set m and ycodes to your data's actual codes.
Trajectory or time axis looks off	tvars not filled with 0..T-1. Rebuild quar with values 0..T-1.
Simulated data has an extra category	cap does not match m. Use cap = m - 1 in %sim_data.
Will not converge	Try fewer classes or a lower deg, provide start_values, or add bounds for stability.